

UNIVERSITY OF GHANA

DEPARTMENT OF STATISTICS

Second Semester IA 1, 2018/2019

STAT 112: Elementary Probability

Answer all questions in Section A and two questions only from Section B (into the answer booklet)

Calculators may be used.

TIME ALLOWED: 1 HOUR, 45 MINUTES

SECTION A

Answer all questions in this section. (40 Marks)

- A1. Find the probabilities $P(A)$ and $P(B)$ if A and B are independent events such that $P(B) = 2P(A)$ and $P(A \cup B) = 5/8$.
- A2. Prove that ${}^nC_k + {}^nC_{k-1} = {}^{n+1}C_k$
- A3. Let X be a discrete random variable with p.m.f given by

$$P(x) = \frac{1}{2^x}, \quad x = 1, 2, 3, \dots$$

- (a) Find $F(x)$ and use it to compute $P(X > 4)$
- (b) Find $P(2 < X \leq 5)$
- A4. Suppose that a random variable X takes on one of the values 0, 1 and 2. If for a constant c , $P(X = i) = cP(X = i - 1)$, $i = 1, 2$, find $E(X)$.
- A5. An ice cream seller takes 20 gallons of ice cream in her truck each day. Let X stand for the number of gallons that she sells. The probability is 0.1 that $X = 20$. If she does not sell all 20 gallons, the distribution of X follows a continuous distribution with a p.d.f. of the form

$$f(x) = \begin{cases} cx & 0 < x < 20 \\ 0, & \text{otherwise} \end{cases}$$

Find the constant c .

SECTION B

Answer ANY TWO questions in this section (30 Marks Each).

- B1. (a) A random variable X has the probability mass function below

$$P(X = x) = \begin{cases} kx & x = 2, 4, 6 \\ k(x - 2) & x = 8 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

- (i) Find k
 - (ii) Find $F(5)$
 - (iii) Compute $\text{Var}(3 - 4X)$ to 3 significant figures
- (b) The random variable Y has the probability distribution given in the table below:

y	-1	0	1	2	3
$P(Y = y)$	p	q	0.2	0.15	0.15

Given that $E(Y) = 0.55$, find the values of p and q .

- B2. (a) A Student in University of Ghana is preparing an itinerary for a visit to six cities in Ghana. The distance travelled, and cost of the trip, will depend on the order in which she visits the cities.
- (i) How many different itineraries are possible?
 - (ii) If Cape coast and Takoradi are two of the cities that the student plans to visit, and she selects one of the itineraries at random, what is the probability that she will visit cape coast before Takoradi?
- (b) In an examination, a candidate is required to pass all five different subjects. Find the number of ways a candidate can fail.
- B3. (a) It is reported that 47% of college freshmen earn a degree and graduate within five years. Assume that graduation records show women make up 50% of the students who graduated within five years, but only 45% of the students who did not graduate within five years. The students who had not graduated within five years either dropped out or were still working on their degrees.
- (i) What is the probability that a female student will graduate within five years?
 - (ii) What is the probability that a male student will graduate within five years?
- (b) Suppose A_1, A_2 and B are events such that A_1 and A_2 are mutually exclusive and $P(B) > 0$. Show that $P(A_1 \cup A_2 | B) = P(A_1 | B) + P(A_2 | B)$